

Chapter 00.

**Course Outline and Syllabus –
ENGG1110**

Learning Outcome

Learning Outcome #1

The ability to apply computer programming to solve engineering problems.

Weekly three-hour lectures

- Fundamentals of programming (***C Language***)
 - Variables, flow control, functions, arrays etc.
- Basics of problem solving via programmatic thinking

Weekly one-hour lab sessions

- Applying programming concepts learned in the lectures
- Hands-on practice through lab exercises

Learning Outcome

Learning Outcome #2

The ability to model a system on a computer to meet specifications.

Related learning activity: Course Project.

- Project specification and requirement to be released
- Problem solving: solution formulation and program testing

Assessment

Overall Assessment Scheme

Participation (7%) and ICAC Talk (to be held <u>31 Oct</u> Tentative) (3%)	10%
Lab Exercises	25%
Midterm Examination (to be held <u>7 Oct</u> Tentative)	10%
Programming Project (individual)	15%
Final Examination Important: score <u>at least 40 out of 100 marks</u> in the final exam in order to PASS this course.	40%
Total	100%

In order to pass this course,
your contributions in ALL FIVE assessment items are required!

Assessment scheme may change due to University arrangements concerning the pandemic

Participation + ICAC Talk (10%)

- To encourage you to participate in the course's learning activities properly, particularly the lab
- **Easiest marks to get!**
- Exact format differs from section to section.
- **PLEASE COME TO LECTURES AND LABS** regardless of your programming background! It's a great way to meet new friends and enjoy university life.
- Starts on Week 3

ICAC Talk (3%)

- ENGG1110 Students are **REQUIRED** to attend an **ICAC Talk**.
- Officials from the Independent Commission Against Corruption will come and talk about **fighting corruption**.
 - As an engineering student and future professional, you should know and understand this important matter!
- Tentative schedule: **31 Oct 2024 (Thu)** during Lecture hours
 - May be in another lecture hall, to be announced later

Lab Exercises (25%)

- Practice programming so that eventually you can tackle the project and the final exam
 - From our experience, if you just sit in the lecture, you won't see much improvement in programming
- You are encouraged to **discuss** with peers, BUT **NOT copying or sharing** C source code.

A sizable portion of midterm/exam questions will be based on the Lab Exercises, so please make sure you do it by yourself properly!

Lab Exercises (25%)

- Usually released every week **on Thu** and due on **Fri 23:59** in the same week
 - Unless otherwise specified
- Most of you will not be able to finish the lab within your section's lab hour. You are encouraged use our **Consultation Hours** if you need further assistance

TA Consultation Hours

- Some of you may need more time and help to finish each week's Lab Exercises
- In addition to our formal lab hours, our TAs will provide **Consultation Hours** for you to ask questions
- Please refer to Blackboard for more information later

Course Schedule

Week #		Lecture Content (Tentative)
1	2 Sep	Introduction, C Programming Basics
2		Data Types and Operators
3		Branching Statements, Basic Debugging
4		Arrays and Characters
5		Basic Looping (while loop)
6	7 Oct	Midterm (Tentative 7 Oct in-class), Debugging looping, For-loop
7		Nested Looping

Course Schedule

Week #		Lecture Content
8	21 Oct	Functions, Variable Scope
9		Project Introduction, lecture (tentative) , Advanced Function Concepts, ICAC Talk in Thursday
10	4 Nov	String Manipulation, , File IO
11		Struct, Congregation on Thu
12		Recursion, Pointer
13		Dynamic Memory Allocation, Conclusion and Revision (End of teaching)
	2 Dec	Reserved for special make-up lecture/labs

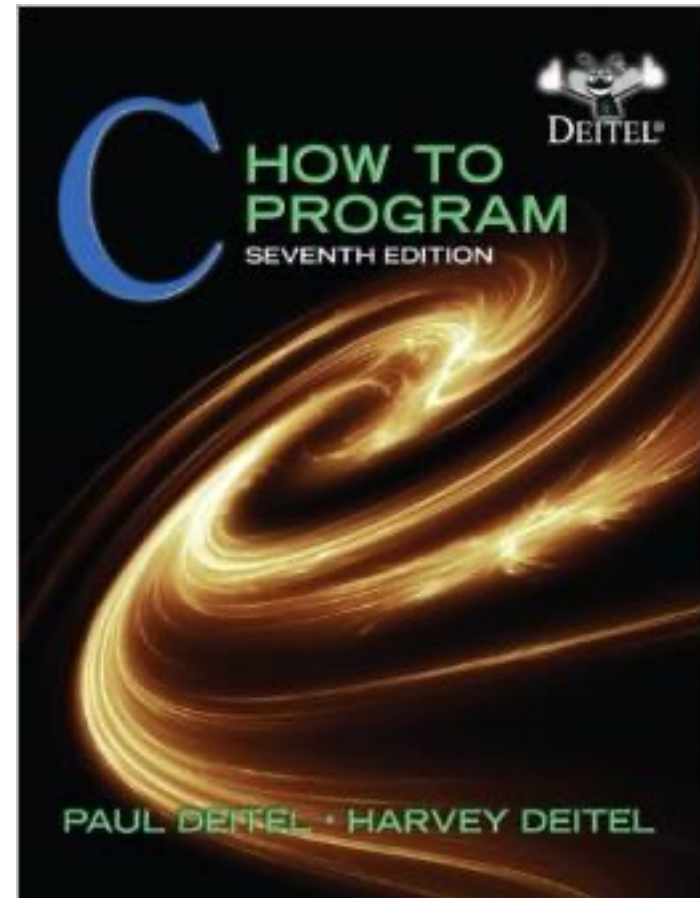
Course Resources

CUHK eLearning Platform

- <https://blackboard.cuhk.edu.hk/>
 - **BlackBoard** course (Web and App)
 - Each section will have their corresponding entry.
- You can find the following:
 - Lecture notes
 - Project materials
 - Announcements
 - Resources and instructions

Textbook and Readings

- Lecture presentation slides are the main teaching materials.
- **Reading Text**
 - C How to Program,**
by Deitel & Deitel (any edition)
 - 7th or 8th ed.
 - Available in CUHK library.
 - Stock in YIA Bookstore.



Online eBook and Micro-modules

- If you prefer an eBook, we have an interactive C Programming **eBook**, with **micro-modules** and online self-learning exercises.
 - **[VPN Required]** <http://cprog.org.cuhk.edu.hk/>
 - By Dr. T. Y. WONG, Mole (CSE, CUHK).
 - The materials are divided into chapters and sections.
 - Micro-modules are short video clips for self-learning.
 - You may attempt the online exercises for self-testing and getting instant feedback.

What VPN?

- When you want to access certain CUHK resources while you are **NOT directly connected to the CUHK** network e.g.
 - via Internet Service Provider (ISP) at home,
 - using mobile data services such as 4G, 5G, WiFi.
- You need CUHK Virtual Private Network (**VPN**)!
 - <https://www.itsc.cuhk.edu.hk/en-gb/all-it/wifi-and-network/cuhk-vpn>
 - **2FA via Duo Mobile** authentication required.
 - Set up once and store your CWEM password.
 - Good idea to set it up at the start of semester!
 - ITSC is improving the connectivity from mainland, so please be patient

Academic Honesty

- **NO PLAGIARISM 嚴禁抄功課**

- No copying from others
- No “*lending*” your work to others
- No employing or using services provided by a third party to undertake submitted work
- No providing services as a third party involving submitted work
- <http://theory.stanford.edu/~aiken/moss/>
(for your eyes only)

- You are prohibited from using any AI tools to complete your assignments, assessments and any other works that count towards the final grade of the course or attainment of the desired learning outcomes
- Every suspected offending case will be reported to the Faculty.
- Must Read: **Honesty in Academic Work**
 - <http://www.cuhk.edu.hk/policy/academichonesty> (CUHK)
 - <https://www.erg.cuhk.edu.hk/erg/AcademicHonesty> (ERG)

IMPORTANT: BYOD

- **Bring Your Own Device** during classes.
- During **lecture + lab combo** session, exercises are carried in-class with students' own notebook computer.
- *Tablet users* should equip with physical keyboard and pointing devices.
- The same requirements as DLCT courses in term 1.
- You may use **Microsoft VM** issued in DLCT course.
- Classroom **CUHK1x WiFi** access needed.